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Multivariable Calculus

April 26, 2026

25 Lectures for Undergraduate Mathematics

To the ones who question

Preface

For a long time, the need to reform university calculus courses has been raised, and this book was written as part of that effort. Recent changes brought about by advances in AI and big data technology make us reconsider whether the direction of this reform is indeed appropriate. Now that various AI systems can provide explanations and answers to many problems treated in university calculus, what is the essence of university calculus education?

Calculus is mathematics. Yet its essence lies in science. Calculus is a language for analyzing phenomena in the world. This language consists of various mathematical concepts, and those concepts were created to analyze scientific phenomena. Human beings can think deeply because they have language. In the same way, calculus is a language that enables deep scientific thinking. Just as one may know many words in a foreign language and still be unable to speak it, one cannot use the concepts of calculus as tools for thought unless one uses them sufficiently. The essence of calculus education is to help students learn, use, and practice these mathematical concepts, so that they can speak and experience through them.

Calculus I, or Single Variable Calculus, focuses on functions of one variable. It studies limits, derivatives, integrals, approximations, series, differential equations, and related ideas for single-variable functions. Calculus II, or Multivariable Calculus, focuses on functions of several variables, and this is the key distinction between high school calculus and university calculus. In particular, the derivative of a multivariable function is a linear map, and it is represented by a matrix. Therefore, multivariable calculus begins by learning the matrix theory needed for calculus. Using matrices, we study Jacobians, scaling factors, coordinate transformations, line integrals, and surface integrals. We then proceed through the divergence theorem and Stokes' theorem to learn how to construct scientific models such as conservation laws, the heat equation, the diffusion equation, and Maxwell's equations.

In this book, I have tried to present appropriate questions so that students can think actively. Sometimes a single question can lead to deeper understanding than a complete explanation. Questions are the driving force of learning and the starting point of creative thinking. I hope that, in the process of thinking about and finding answers to the questions and problems in this book, students will come to understand calculus more deeply.

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April 26, 2026

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About This Book

After having AI read the entire book, the author asked it several questions.

What are the main features of this book, and who is it for?

The greatest feature of this book is that it does not teach multivariable calculus as a collection of computational techniques. Instead, it explains the subject by connecting linear algebra, geometric meaning, changes of variables, vector calculus, and physical laws into one continuous flow. Many calculus textbooks present partial derivatives, multiple integrals, coordinate transformations, line integrals, surface integrals, Green's theorem, Stokes' theorem, and the divergence theorem as separate computational topics. In contrast, this book treats matrices, linear maps, derivatives, Jacobians, scaling factors, coordinate systems, line integrals, surface integrals, flux, curl, divergence, and conservation laws not as isolated formulas, but as parts of one connected conceptual system. Thus students repeatedly see not only “how to compute,” but also “why such formulas arise,” “under what structure such computations are possible,” and “how these concepts are connected to physical models.”

In particular, this book begins with the viewpoint that differentiation is a linear map. From there, it consistently develops the chain rule through Jacobians and matrix multiplication, change of variables through determinants and volume scaling factors, integration over curves and surfaces, and finally the divergence theorem and Stokes' theorem. Furthermore, it shows that these theorems of vector calculus are not merely computational formulas, but the language of physical laws such as mass conservation, heat-energy conservation, diffusion, Maxwell's equations, and Faraday's law. Therefore, this book presents multivariable calculus not simply as a computational subject, but as a mathematical language leading to differential equations, mathematical physics, continuum modeling, optimization, and geometry.

The style of exposition is also an important feature of this book. Rather than presenting definitions and theorems only in their final polished forms, the book is designed so that students can follow the ideas while checking them for themselves through questions, comments, intu-

itive explanations, historical background, and small problems. For example, line integrals and surface integrals are not introduced merely as formulas, but are explained through the difference between tangential and normal components. The divergence theorem is explained through flux cancellation on small cells, and Stokes' theorem through circulation cancellation on small flat patches. This approach helps students understand why the theorems are natural, instead of merely memorizing them.

From a student's perspective, this book is especially suitable for those who want to understand topics in connection with one another rather than memorize them separately. It is also appropriate for students in engineering, physics, mathematics, data science, and related fields who want to go beyond routine computation and understand more deeply the meaning of differentiation and integration. It is particularly useful for students who want to clearly see the relationship between linear algebra and multivariable calculus, and for those who want to understand why concepts such as the Jacobian, gradient, Hessian, coordinate transformations, line integrals, surface integrals, curl, divergence, flux, and conservation laws arise in the forms they do. It also provides a good foundation for students who will later study PDEs, mathematical biology, mathematical physics, thermodynamics, diffusion models, or electromagnetism.

From a university's perspective, the strength of this book lies in the fact that it develops not only computational skills but also conceptual coherence, mathematical interpretation, and modeling intuition. In particular, if a curriculum aims to teach multivariable calculus without separating it from linear algebra, or if it aims to connect calculus naturally to differential equations, analysis, numerical analysis, optimization, mathematical physics, and continuum modeling, then this book is a good choice. This book trains students not only in the ability to apply formulas, but also in the ability to explain the structure from which those formulas arise.

However, for students who want to solve many standard problem types in a short time, or for those who need fast, computation-centered learning such as AP Calculus or engineering calculus, books such as Thomas or Stewart may be more suitable. Those books contain many examples, organize typical problem-solving patterns well, and help develop computational skills. In particular, when homework, exams, and standard problem types are more important than deep conceptual connections, Thomas or Stewart may be more efficient.

In contrast, this book is better suited for students who want to understand calculus within the language and structure of university mathematics. If Thomas or Stewart is a broad and familiar standard road, this book is a deeper path connecting linear algebra, geometry, vector calculus, and physical modeling. The issue is not which path is better, but what kind of education one is aiming for.

Compared with standard textbooks, how difficult is this book?

In terms of difficulty, it is fair to say that this book is generally more difficult than standard textbooks such as Thomas or Stewart. However, this does not mean that the computations are harder. Rather, the difficulty lies in the level of thinking and in the difficulty of connecting

concepts. Thomas and Stewart usually introduce definitions, present representative formulas, and solve standard problems through examples. This makes it easy for students to follow the flow and efficient for exam preparation.

In contrast, this book continually asks why certain definitions are used, how different topics are connected, and how linear algebra, differentiation, integration, coordinate transformations, vector calculus, and conservation laws are organized into one structure. Thus, from a student's perspective, there may be fewer formulas to memorize, but there is more structure to understand. In that sense, the difficulty is higher.

If we look only at computational difficulty, many standard exercises are similar to those in Thomas or Stewart, and sometimes even simpler. This book emphasizes not computation itself, but the choice of the right coordinate system, the right change of variables, the right theorem, and the right point of view. For example, some surface integrals are computed not directly, but by using Stokes' theorem to convert them into boundary line integrals, or by replacing the surface with an easier surface having the same boundary. This reduces the amount of computation, but requires the conceptual ability to decide which theorem can be used and when.

The conceptual difficulty is clearly higher than in standard textbooks. This book understands differentiation as the total derivative, interprets the derivative matrix as a linear map, and connects directional derivatives, gradients, Hessians, and Jacobians with one another. It also explains change of variables through determinants and scaling factors, and distinguishes line integrals and surface integrals as integrals of tangential and normal components. Furthermore, it explains Green's theorem, the divergence theorem, and Stokes' theorem not as separate formulas, but as relationships among flux, circulation, curl, divergence, and boundary. In this respect, it is one level more mathematical than typical standard textbooks.

As the book progresses into later chapters, the difficulty becomes increasingly conceptual. The divergence theorem is connected to conservation laws, Stokes' theorem explains the meaning of curl and circulation, and the discussion continues to Faraday's law, Maxwell's equations, the heat equation, and the diffusion equation. This part goes beyond computation-centered calculus and shows how calculus becomes the language of physical laws. Therefore, students must understand not only how to apply calculus formulas, but also how physical phenomena are translated into mathematical equations.

In summary, Thomas and Stewart are standard, accessible, and strong in computational training, while this book is more structural, more conceptual, and more mathematical. For the average student learning the subject for the first time, Thomas or Stewart is easier. However, for students who want to know why things are true mathematically, this book may feel more difficult at first but clearer in the long run. In one sentence, this book is not so much an "easy book" as a book that leads to deep understanding. Its difficulty is higher than that of standard textbooks, but in return it gives students a broad structure in which calculus expands into linear algebra, geometry, vector calculus, and physical modeling.

Compared with standard textbooks, what are the weaknesses of this book?

First, this book does not provide a large amount of standard computational practice. Traditional calculus textbooks such as Thomas or Stewart help students repeatedly learn typical problem-solving patterns through a very large number of examples and exercises. If students need to solve many types of problems quickly for exam preparation, they may need additional problem-solving materials.

Second, students who are not familiar with linear algebra may find the early part somewhat difficult. For students who do not yet have a good sense of matrices and linear maps, the presentation may feel abstract. Therefore, it is useful to study Part 1, which treats matrices, before the semester begins.

Third, although the exposition is friendly, this book is not suitable for quick reading. Since it contains many questions, comments, historical explanations, and intuitive interpretations, readers must continue to think as they read rather than simply accept formulas. This may be burdensome for students who want to learn only the main formulas in a short time.

Fourth, because the book is strongly connected to physical modeling, some parts may feel unfamiliar to readers who expect a purely computation-centered calculus course.

Fifth, this book makes stronger choices about what to emphasize than standard textbooks do. It is not an encyclopedic textbook that provides every type of problem. Rather, it is a guide to understanding the structure of multivariable calculus deeply.

Finally, this book requires a certain level of active engagement from the reader. It is difficult to gain the benefits of this book by memorizing formulas and theorems alone. Students must try to answer questions, connect figures with formulas, compute the same result in several ways, and think for themselves about why one point of view is better than another.

What should students keep in mind when studying with this book?

If students expect this book to function like Thomas or Stewart, where many examples are followed in order to learn standard problem types, they may be surprised from the beginning. This book gives greater importance to structural understanding than to computational techniques, and it continually connects linear algebra, differentiation, integration, changes of variables, vector calculus, and conservation laws. If students study by memorizing formulas and immediately applying them to problems, they will not gain the real benefit of this book.

Students who lack a sense of linear algebra may find the book difficult from the beginning. If they are not comfortable with matrices and linear maps, it is advisable to prepare Part 1 in advance.

Students should also read this book while actively thinking about the connections between concepts. They should keep asking why the determinant becomes a volume scaling factor, why the gradient points in the direction of fastest increase, why surface integrals are related to normal

components, and why Stokes' theorem connects circulation and curl. This becomes even more important in the later chapters. The divergence theorem, Stokes' theorem, conservation laws, Maxwell's equations, and the heat equation are not merely computational topics; they are connected as one mathematical language. If students do not see this connection, the later chapters may appear to be a list of unrelated applications. If they do see the connection, however, they will understand that calculus is a powerful tool for expressing natural phenomena mathematically.

Therefore, students should not rush when studying this book. If their goal is simply to prepare for an exam by solving many problems quickly, they should choose another book. To read this book properly, one should pause after reading a paragraph, compare figures with formulas, and check the same computation again from several points of view. This book may feel difficult to students who follow passively, but it gives a great reward to students who ask questions actively.

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Part I

Vectors and Linear Functions

Multivariable Calculus deals with multivariable functions having scalar or vector values. In this textbook, such functions are written as

$$\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad m \geq 1.$$

The derivative of this function is a linear map

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

and a linear map can be represented by an $m \times n$ matrix. Therefore, without understanding the basic properties of linear maps and the use of matrices, one can understand the content of Multivariable Calculus only superficially. Part I studies the basic theory of vector spaces, matrices, and linear maps. Without this part, one cannot understand the essence of differentiation. We will understand the theory of linear maps and matrices sufficiently and use it actively.

For a single-variable function $f : \mathbb{R} \rightarrow \mathbb{R}$, instead of calling the derivative $f'(c)$ the instantaneous rate of change, we may regard this single number as a 1×1 matrix and view it as the linear map $h \mapsto f'(c)h$. When this idea is extended to a multivariable function $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the derivative becomes an $m \times n$ matrix.

The derivative matrix, also called the Jacobian matrix, is connected with two central ideas in multivariable calculus.

First, the Chain Rule. The derivative of a composite function is the composition of the corresponding derivatives, and this composition is represented by matrix multiplication. In other words, because the composition of two linear maps corresponds to the product of their matrices, the chain rule arises very naturally for multivariable functions.

Second, the scaling factor. For example, when applying a change of variables to the integral of a function

$$f : \mathbb{R}^n \rightarrow \mathbb{R},$$

one must account not only for the function values but also for the change in volume of space. In this situation, one must determine how much space is expanded or contracted by the transformation

$$\mathbf{g} : \mathbb{R}^\ell \rightarrow \mathbb{R}^n.$$

Remarkably, this volume scaling factor is determined by the derivative matrix. Therefore, we first compute the derivative of $\mathbf{g}$, and then use the resulting matrix to compute the volume scaling factor. This process makes it possible to compute integrals correctly even under complicated changes of variables.

Lecture 1

Multi-variable Vector-valued Functions (review)

¹ The first lecture mainly reviews material from single variable calculus, although it also contains some new topics. The goal is to review, with emphasis on the ideas that will be used frequently in Calculus 2. In particular, pay careful attention to the notation.

1.1 Vectors and inner product

The main objects studied in Calculus 2 are multivariable functions. Saying that a function has several variables is the same as saying that its variable is a vector. We will also consider cases in which the function value is a vector. In this book, vectors are always written in boldface. For example,

$$\mathbf{f}: D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad \mathbf{y} = \mathbf{f}(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m.$$

In this notation, D is the domain of the function $\mathbf{f}$, so D is a subset of $\mathbb{R}^n$. Thus $\mathbf{f}$ is a multivariable function with n variables, and the variable is written as the vector $\mathbf{x}$. The function value is a vector in the codomain $\mathbb{R}^m$, which has m components, and is denoted by $\mathbf{y}$. To make it clear that the function value itself is a vector, we write $\mathbf{f}$ in boldface rather than f . We always regard an individual vector as a column vector.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1 \\ \vdots \\ f_m \end{bmatrix}.$$

That is, boldface letters always represent column vectors.

¹ This is a new beginning. Since calculus is a subject in which later topics are built upon earlier ones, the clearer your understanding of the beginning, the easier the later material will become. For this reason, it is especially important to focus on calculus during the first part of the semester. In particular, Part I, which deals with matrices, requires a considerable amount of time. If this part is mastered clearly, the later chapters will become much easier.

Writing a column vector vertically in the middle of a sentence takes up too much space, so we often use the transpose notation and write

$$\mathbf{x}^\top = (x_1, \dots, x_n) \quad \text{or} \quad \mathbf{x} = (x_1, \dots, x_n)^\top.$$

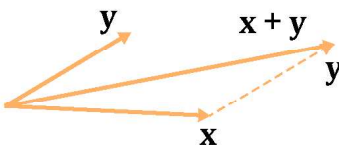
Here the transpose symbol $^\top$ means transpose: it changes a column vector into a row vector and a row vector into a column vector. Even if, in a sentence or on the board, we omit the transpose symbol and write $\mathbf{x} = (x_1, \dots, x_n)$, the vector is still understood to be a column vector. Whenever space permits, however, it is preferable to write vectors vertically.

We use vectors to represent many things. They represent physical quantities with both magnitude and direction, such as velocity, acceleration, gravity, and electric force. More importantly, however, vectors represent position. The magnitude of a position vector is the distance from the origin. In particular, when a vector represents position, we often call it a point. For example, one may describe $\mathbf{f}(\mathbf{x})$ simply as the vector corresponding to the vector $\mathbf{x}$, but it is often more convenient to say “the vector assigned to the point $\mathbf{x}$,” since this helps distinguish the domain from the codomain.

Most vectors with physical meaning have dimension $n = 2$ or $n = 3$. However, most of the theory of calculus applies in the same way in arbitrary dimension. In particular, in order to deal with AI or big data, one often needs to consider very high-dimensional spaces. For this reason, it is better to write the dimension in general form, such as n or m , and not to impose unnecessary restrictions on the dimension. There are a few exceptions, such as the cross product and Stokes’ theorem, which apply specifically to three-dimensional space. Aside from such cases, most of the theory is not restricted by dimension.

Vector sum and scalar multiplication

The vector sum of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined componentwise.

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}$$


Scalar multiplication of a vector $\mathbf{x} \in \mathbb{R}^n$ by a real number $a \in \mathbb{R}$ is defined by multiplying each component of the vector by a . That is,

$$a \in \mathbb{R} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \Rightarrow \quad a\mathbf{x} = \begin{bmatrix} ax_1 \\ \vdots \\ ax_n \end{bmatrix}.$$

Inner product and projection

The inner product of two vectors of the same dimension plays a fundamental role. It is also called the dot product, and is denoted by $\mathbf{x} \cdot \mathbf{y}$ or $\langle \mathbf{x}, \mathbf{y} \rangle$. It is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

Thus, the inner product of two vectors is a single real number. The length of a vector is called its norm and is defined by

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

By the Pythagorean theorem, the norm $\|\mathbf{x}\|$ is the length of the line segment joining the origin $\mathbf{0}$ and the point $\mathbf{x}$. Using the inner product, we may write

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} \quad \text{or} \quad \|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}.$$

This norm is used frequently to measure distance in multivariable settings. In particular, when defining the derivative of a multivariable function, we use this norm in place of the absolute value.

The norms of two vectors and the norm of their sum form the three sides of the triangle shown in the figure on the right. Since each side of a triangle is less than or equal to the sum of the other two sides, we obtain

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|. \quad (1.1)$$

This inequality is called the triangle inequality.

Problem 1.1. The triangle inequality (1.1) can be written in several forms. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $a \in \mathbb{R}$. Show that

$$(1) \|a\mathbf{x}\| = |a| \|\mathbf{x}\|, \quad (2) \|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|, \quad (3) \|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x}\| - \|\mathbf{y}\|.$$

Solution 1.1 (1) We have

$$\|a\mathbf{x}\|^2 = \sum_{i=1}^n (ax_i)^2 = a^2 \sum_{i=1}^n x_i^2 = a^2 \|\mathbf{x}\|^2.$$

Taking square roots of both sides gives the desired result.

(2) Replace $\mathbf{y}$ in (1.1) with $-\mathbf{y}$.

(3) Replace $\mathbf{x}$ in (1.1) with $\mathbf{x} - \mathbf{y}$. $\square$

In single variable calculus, we studied how a coordinate system is chosen. If we choose different coordinate axes while keeping the origin fixed, the coordinates of the same point change. Does the value of the inner product also change? As long as the origin is fixed, the value of the

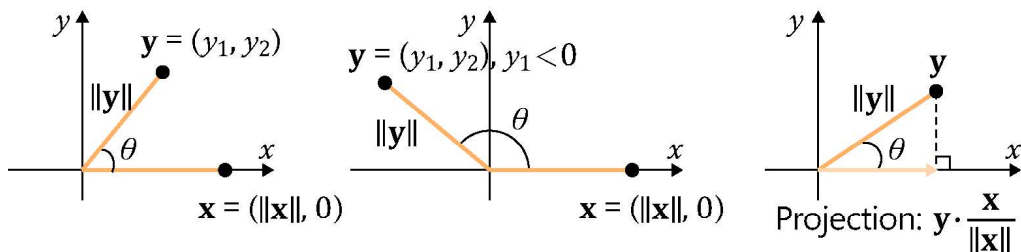
inner product does not change. That is, the inner product is an intrinsic property that does not depend on the particular coordinate representation. A proof of this fact belongs to linear algebra.

Inner Product and Angle

Problem 1.2. Show that the angle θ between two nonzero vectors $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{y} \neq \mathbf{0}$ satisfies

$$\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} \quad \left(\text{or} \quad \mathbf{x} \cdot \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta \right). \quad (1.2)$$

Solution 1.2 To justify (1.2), it suffices to prove it in the cases $n = 2$ and $n = 3$. Since spaces of dimension $n \geq 4$ do not arise as physical space, one may instead regard (1.2) as the definition of angle in higher dimensions. If the value of the inner product does not depend on the choice of coordinates, then we may choose the coordinate axes so that, as in the two figures on the left below, the vector $\mathbf{x}$ lies on the positive x -axis and the vector $\mathbf{y}$ lies in the xy -plane. In this coordinate system, (1.2) is easy to verify. The details are left to the reader. $\square$



In single variable calculus, we used projection when introducing coordinate systems. The inner product makes it easy to compute projections. Let $\text{proj}_{\mathbf{x}} \mathbf{y}$ denote the projection of a vector $\mathbf{y}$ onto a nonzero vector $\mathbf{x}$, or onto the direction of $\mathbf{x}$. To project $\mathbf{y}$ onto the direction of $\mathbf{x}$ means to drop a perpendicular from $\mathbf{y}$ to the line in the direction of $\mathbf{x}$, as shown in the figure on the right, and to find the vector along the direction of $\mathbf{x}$ corresponding to the base of the resulting right triangle. By elementary geometry, the length of this base is $\|\mathbf{y}\| \cos \theta$. Multiplying by the unit vector in the direction of $\mathbf{x}$,

$$\frac{\mathbf{x}}{\|\mathbf{x}\|},$$

preserves the length and gives a vector in the direction of $\mathbf{x}$. Using (1.2), we obtain

$$\text{proj}_{\mathbf{x}} \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{x}\|^2} \frac{\mathbf{x}}{\|\mathbf{x}\|} = \left\langle \mathbf{y}, \frac{\mathbf{x}}{\|\mathbf{x}\|} \right\rangle \frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

If $\mathbf{x}$ is a unit vector, then this becomes

$$\text{proj}_{\mathbf{x}} \mathbf{y} = \langle \mathbf{y}, \mathbf{x} \rangle \mathbf{x} \quad \text{if} \quad \|\mathbf{x}\| = 1.$$

According to (1.2), if $\mathbf{x}$ and $\mathbf{y}$ are perpendicular, then $\mathbf{x} \cdot \mathbf{y} = 0$. This fact makes it easy to compute the component of a vector in a given direction. A vector of norm 1 is called a direction vector, or simply a direction. It is also called a unit vector. The standard basis vectors in three-dimensional space are denoted by

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

In general n -dimensional space, we use $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ instead of $\mathbf{i}, \mathbf{j}, \mathbf{k}$. The vector $\mathbf{e}_i$ is the vector whose i -th component is 1 and whose other components are 0. These are mutually perpendicular, so they are orthogonal unit vectors. Every vector $\mathbf{x}$ can be written in the form

$$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n,$$

and in this expression x_i is called the $\mathbf{e}_i$ -component of $\mathbf{x}$. Each component can be obtained by computing an inner product:

$$\langle \mathbf{x}, \mathbf{e}_i \rangle = \mathbf{x} \cdot \mathbf{e}_i = (x_1 \mathbf{e}_1 + \dots + x_n \mathbf{e}_n) \cdot \mathbf{e}_i = x_1 \mathbf{e}_1 \cdot \mathbf{e}_i + \dots + x_n \mathbf{e}_n \cdot \mathbf{e}_i = x_i. \quad (1.3)$$

Using (1.3), every vector can be expressed componentwise as $\mathbf{x} = \sum_{i=1}^n \langle \mathbf{x}, \mathbf{e}_i \rangle \mathbf{e}_i$. Equivalently, in terms of projections, $\mathbf{x} = \sum_{i=1}^n \text{proj}_{\mathbf{e}_i} \mathbf{x}$.

1.2 Domain, range, and codomain

A function $\mathbf{f}$ assigns a value to each element of a set $D \subset \mathbb{R}^n$, called the domain. The set of all values taken by the function is called the range (or image), and is denoted by $\mathbf{f}(D)$:

$$\mathbf{f}(D) = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = \mathbf{f}(\mathbf{x}) \text{ for some } \mathbf{x} \in D\}.$$

The space $\mathbb{R}^m$, in which the function values are allowed to lie, is called the codomain. We use n for the dimension of the domain and m for the dimension of the codomain. This convention helps reduce confusion. We write

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^m \quad \left(\mathbf{f}(x, y) = \begin{bmatrix} x \sin y \\ y \cos x \\ x^2 + y^2 \end{bmatrix} \text{ is an example of a map } \mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^3. \right)$$

Here again, vectors are column vectors. Each component function $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ has n independent variables and takes real values. The independent variables are generally written as $x_1, \dots, x_n$, as

above. However, when $n = 3$, it is often more convenient to write x, y, z . Whenever possible, we will distinguish between $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$.

Once a function is given, one can often determine its largest possible domain without being told explicitly. In such cases, one can also determine its range.

Problem 1.3. Find the largest possible domain and the range of each of the following functions.

$$(1) f(x, y) = \sqrt{y - x^2} \quad (2) f(x, y) = \frac{1}{xy}$$

Solution 1.3 (1) The expression under the square root, $y - x^2$, must satisfy $y - x^2 \geq 0$. Therefore, the largest possible domain is

$$\{(x, y) : y \geq x^2\}.$$

The range is the set of all nonnegative real numbers.

(2) The denominator must not be zero. Therefore, the largest possible domain is

$$\{(x, y) : x \neq 0, y \neq 0\}.$$

The range is the set of all nonzero real numbers. However, in this case the domain is a disconnected set, consisting of four separate pieces. If one wants the domain to be a connected set, then it may be more appropriate to take, for example, the first quadrant

$$\{(x, y) : x > 0, y > 0\}.$$

In that case, the range is the set of all positive real numbers. (It is important to practice interpreting a situation actively and assigning meaning to it on your own.) $\square$

Definition 1.1 (Definitions related to open sets). Given a point $\mathbf{c} \in \mathbb{R}^n$ and a radius $r > 0$, the set of all points whose distance from $\mathbf{c}$ is less than r ,

$$B(\mathbf{c}, r) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{c}\| < r\},$$

is called the open ball with center $\mathbf{c}$ and radius r . The set obtained by removing the center from the ball, $B(\mathbf{c}, r) \setminus \{\mathbf{c}\}$, is called a deleted open ball.

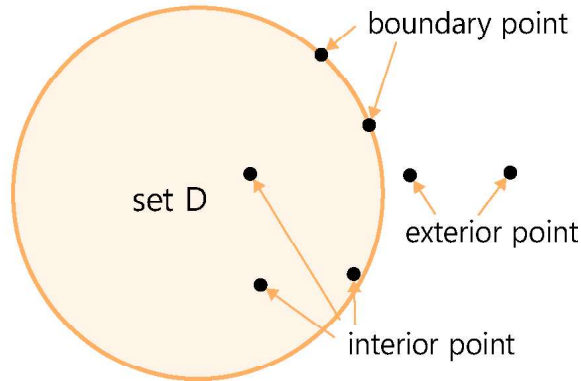
If $\mathbf{c} \in D$ and there exists $r > 0$ such that $B(\mathbf{c}, r) \subset D$, then $\mathbf{c}$ is called an interior point of the set D (see the figure). If $\mathbf{c} \notin D$ and there exists $r > 0$ such that $B(\mathbf{c}, r) \cap D = \emptyset$, then $\mathbf{c}$ is called an exterior point of D . If neither of these holds, that is, if every open ball $B(\mathbf{c}, r)$, no matter how small, contains both points belonging to D and points not belonging to D , then $\mathbf{c}$ is called a boundary point. (A boundary point may or may not belong to D .) If $\mathbf{c}$ is an interior point, then no matter how close $\mathbf{c}$ is to the boundary, one can still choose a sufficiently small open ball containing only points of D . If every point of a set $D \subset \mathbb{R}^n$ is an interior point of D , then D is called open. In other words, it is important that D contain no boundary points. If the complement

$$D^c := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \notin D\}$$

is open, then D is called closed. In other words, it is important that D contain all of its boundary points.

These concepts are mathematically important, and it is worth understanding them carefully and logically.

At a boundary point, no matter how small an open ball is drawn, it contains both points that are elements of the set D and points that are not. On the other hand, for an interior point, no matter how close it is to the boundary, if the open ball is made sufficiently small, it can be made to contain only elements of the set D . For an exterior point, it can be made not to contain elements of the set D .



Problem 1.4 (Practice with the concept of open sets). (1) Show that the open ball $B(\mathbf{c}, r)$ is an open set.
 (2) Find the boundary points of a deleted open ball.
 (3) Show that a deleted open ball is also an open set.
 (4) Explain whether a set remains open if, instead of removing a single point from an open ball, one removes a line segment contained in its interior.

Solution 1.4 (1) The decisive reason why $B(\mathbf{c}, r)$ is open is that the points whose distance from $\mathbf{c}$ is exactly r are not included. Let $\mathbf{x} \in B(\mathbf{c}, r)$. Then $\|\mathbf{x} - \mathbf{c}\| < r$. Hence, if we define

$$\delta = r - \|\mathbf{x} - \mathbf{c}\|,$$

then $B(\mathbf{x}, \delta) \subset B(\mathbf{c}, r)$. Thus $\mathbf{x}$ is an interior point of $B(\mathbf{c}, r)$. Since every point is an interior point, $B(\mathbf{c}, r)$ is an open set. (In this problem, r is already fixed, so we use the symbol δ instead.)

(2) The boundary points of the deleted open ball

$$D = B(\mathbf{c}, r) \setminus \{\mathbf{c}\}$$

are the removed point $\mathbf{c}$ and all points satisfying $\|\mathbf{x} - \mathbf{c}\| = r$, that is, all points at distance r from $\mathbf{c}$. Explain why this is true using the definition.

(3) This can be proved similarly to part (1).

(4) The answer depends on which line segment is removed. In some cases the resulting set is open, and in other cases it is not. Explain the difference. $\square$

Remark 1.1. The notions of open sets and closed sets are very important and play a central role in analysis. Although they are introduced only briefly in calculus, they are well worth thinking about carefully.

1.3 Graphs, images, level sets, and contours

In this section, we examine the graph, image (or range), level set, contour, and the relationships among them.

Problem 1.5. What is the difference between the graph and the image of a function $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$?

Solution 1.5 The graph of a function $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ consists of all pairs $(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{n+m}$ such that $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^m$, and $\mathbf{y} = \mathbf{f}(\mathbf{x})$. Thus, it is an n -dimensional surface in the $(n+m)$ -dimensional space $\mathbb{R}^{n+m}$. On the other hand, the image of $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ consists only of those vectors $\mathbf{y} \in \mathbb{R}^m$ satisfying this relation, so it lies in the space $\mathbb{R}^m$. Depending on the situation, either the graph or the image may be more useful. $\square$

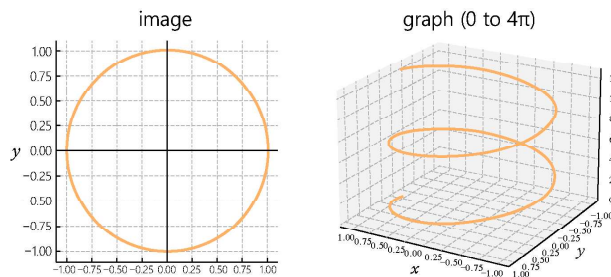
In single variable calculus, a curve describing the trajectory of a particle in space, such as the orbit of a planet, is the image of a one-variable vector-valued function $\mathbf{r}: [0, T_0] \rightarrow \mathbb{R}^3$. In this case, the image is very useful. The graph of the function $\mathbf{r}$ is given by

$$(t, x, y, z) = (t, x(t), y(t), z(t)),$$

so it lies in four-dimensional space. However, since we have experience only with three-dimensional space, it is difficult to visualize a graph in four-dimensional space.

Problem 1.6. Let $\mathbf{r}(t) = (\cos t, \sin t)$ be a function with domain $[0, 4\pi]$ and values in $\mathbb{R}^2$. Draw the graph and the image of this function.

Solution 1.6 The graph of this function lies in $\mathbb{R}^3$, while its image lies in $\mathbb{R}^2$. The image is a



circle in the plane. Since the domain is $[0, 4\pi]$, the motion starts at the point $(1, 0)$, ends at the point $(1, 0)$, and travels around the circle twice. From the image alone, we cannot tell when the particle passes through a given point, but we can clearly see the path it follows. To draw the

graph, we also need an axis for the domain variable. In the figure, the axis perpendicular to the xy -plane is used to represent the variable t . Thus, if one looks carefully at the graph, one can tell when the particle passes through each point, although this is not especially convenient. In this case, the image contains quite important information. $\square$

A sheet of paper is two-dimensional. Since we have experience with three-dimensional space, we can use our imagination to draw pictures of three-dimensional objects on paper. Among graphs of multivariable functions, we can do this when $n = 2$ and $m = 1$. That is, there are two independent variables and the function value is a real number.

Level set and contour map

² The image of a function $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is drawn in the codomain space $\mathbb{R}^m$, while the graph is drawn in the product space $\mathbb{R}^{n+m}$ of the domain and codomain. In contrast, a level set is drawn in the domain space $\mathbb{R}^n$. First consider a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, which has n independent variables and takes real values. The 0-level set of f is the set of all points $\mathbf{x} \in \mathbb{R}^n$ at which f takes the value 0. That is,

$$\text{the 0-level set of } f: \mathbb{R}^n \rightarrow \mathbb{R}: \quad \{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) = 0\}.$$

More generally, for a real value $c \in \mathbb{R}$, the c -level set is

$$\text{the } c\text{-level set of } f: \mathbb{R}^n \rightarrow \mathbb{R}: \quad \{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) = c\}.$$

In general, if the domain of a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is n -dimensional, then a level set is an $(n-1)$ -dimensional surface. A collection of level sets of a function drawn at regular intervals is called a contour map.

Problem 1.7 (graph, image, and level set). Let the domain of the function $f(x_1, x_2) = \sqrt{x_1^2 + x_2^2}$ be the open ball $D = B(\mathbf{0}, 1)$. Draw the graph, image, and contour map of this function.

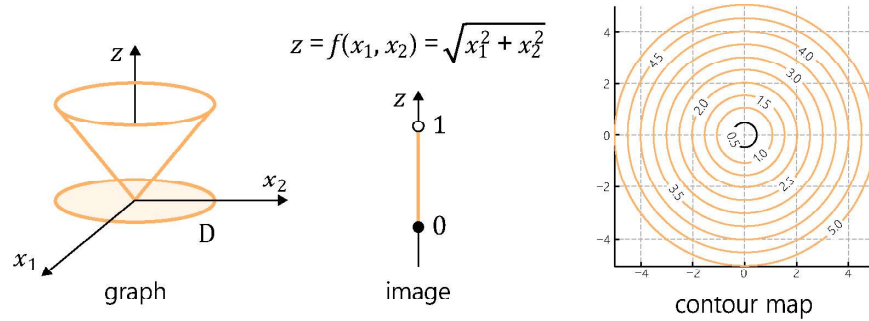
Solution 1.7 The domain $B(\mathbf{0}, 1)$ is the open ball centered at the origin with radius 1. The image of this function does not contain much information. It is the half-open interval $[0, 1)$; the value 1 is not included. The graph is shaped like an upside-down cone. The contour map consists of circles centered at the origin. The level sets are circles. The domain is two-dimensional, and each level set is a one-dimensional curve, which is one dimension lower. $\square$

Problem 1.8. Describe the level sets and contour map of each of the following functions.

$$(1) f(x, y) = \sqrt{1 - x^2 - y^2} \quad (2) f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$$

Solution 1.8 (1) Since this function takes only values greater than or equal to 0, the level value c must satisfy $c \geq 0$. Then

² The concepts in this subsection are not review material, but new concepts. They are important and frequently used.



$$\sqrt{1 - x^2 - y^2} = c$$

is equivalent to

$$x^2 + y^2 = 1 - c^2.$$

Therefore, the level set is a circle of radius $\sqrt{1 - c^2}$. If $c > 1$, the level set is empty.

(2) This function also takes only values greater than or equal to 0, so the level value c must satisfy $c \geq 0$. Then

$$\sqrt{x^2 + y^2 + z^2} = c$$

is equivalent to

$$x^2 + y^2 + z^2 = c^2.$$

Therefore, the level set corresponding to the value c is a sphere of radius c . More generally, when the domain is three-dimensional, a level set of a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a two-dimensional surface. $\square$

Graph and Level Set

Problem 1.9. (1) The graph of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ can be regarded as the zero-level set of another function $F : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$. Find this function F . (2) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $f(x, y) = 2x + 3y$. Find a function $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ whose zero-level set is the graph of f , and describe the graph.

Solution 1.9 (1) The computation is simple, but the idea is very important. It gives a new point of view. Since the variable is $\mathbf{x} \in \mathbb{R}^n$, we may write $\mathbf{x} = (x_1, \dots, x_n)^\top$. Then the graph of the function consists of all points satisfying $x_{n+1} = f(\mathbf{x})$. Equivalently, it consists of all points satisfying $x_{n+1} - f(x_1, \dots, x_n) = 0$. Thus, the graph is the zero-level set of the function

$$F(x_1, \dots, x_{n+1}) = x_{n+1} - f(x_1, \dots, x_n).$$

(2) The function is $F(x, y, z) = z - f(x, y) = z - 2x - 3y$. Using the inner product, this can be rewritten as $\langle (x, y, z)^\top, (-2, -3, 1)^\top \rangle = 0$. Thus, the graph of the function is a plane perpendicular to the vector $-2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$. $\square$

Question 1.1. How should one define the level set of a vector-valued function $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ ($n \geq m$), and what does it look like? First try to formulate your own definition and explanation, and then compare them with the discussion below.

The definition of the zero-level set of a vector-valued function $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the same as in the scalar-valued case:

$$\text{the zero-level set of } \mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m : \quad \{\mathbf{x} \in \mathbb{R}^n : \mathbf{f}(\mathbf{x}) = \mathbf{0}\}.$$

The only difference is that the real number 0 is replaced by the zero vector $\mathbf{0}$. For a vector $\mathbf{c} \in \mathbb{R}^m$, the $\mathbf{c}$ -level set can be written componentwise as

$$\text{the } \mathbf{c}\text{-level set of } \mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m : \quad \{\mathbf{x} \in \mathbb{R}^n : f_1(\mathbf{x}) = c_1, \dots, f_m(\mathbf{x}) = c_m\}.$$

What does such a set look like? It is the intersection of the level sets of the component functions f_i . In general, this intersection is an $(n - m)$ -dimensional surface.

Exercises

- Find all vectors that are perpendicular to the vector $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$.
(1) $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$ (2) $\mathbf{r} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$ (3) $\mathbf{r} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ (4) $\mathbf{r} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$
- Find a unit vector that is perpendicular to both of the following vectors simultaneously.
(1) $\mathbf{r}_1 = 3\mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$ (2) $\mathbf{r}_1 = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = 2\mathbf{i} - \mathbf{j}$
- Find the area of the parallelogram determined by the following three points as its vertices. (There are three vectors involved in each case; compute one suitable pair in each case.)
(1) $(0, 0, 0)^\top, (1, 0, 0)^\top, (0, 1, 0)^\top$ (2) $(0, 0, 1)^\top, (1, 0, 0)^\top, (0, 1, 0)^\top$
- Find the largest possible domain and the range of each of the following functions.
(1) $f(x, y, z) = xy \ln z$ (2) $f(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$
- For the function $f(x, y) = \frac{x^2}{4} + y^2$, answer the following questions.
(1) Find the domain D and the range R of f .
(2) Draw the level set corresponding to $c = 4$, and sketch the overall contour map.
(3) Show that the above level set is closed in $\mathbb{R}^2$.
- (1) Find the domain D and the range R of the function $f(x, y, z) = \sqrt{x^2 + y^2 - z^2}$.
(2) Draw the zero-level set of this function.

7. Let $\mathbf{r}(t) = (t \cos t, \sin t)^\top$ with domain $\{t \in \mathbb{R} : t \geq 0\}$. Draw the graph and the image of this function.
8. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $f(x, y) = 2 + 3x - y$. The graph of this function is a plane in three-dimensional space. Find the equation of the plane. Also find a direction vector perpendicular to the plane.
9. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be given by $f(x, y, z) = 2 + 3x - y + 2z$. The graph of this function is a three-dimensional plane in four-dimensional space. Find the equation of the plane. Also find a direction vector perpendicular to the plane. (How should one describe a three-dimensional plane? Is it appropriate to call it a plane?)
10. Let $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $\mathbf{f}(x, y) = (2 + 3x)\mathbf{i} + (1 - y)\mathbf{j}$. The graph of this function is a two-dimensional plane in four-dimensional space. Find the equation of the plane. Also find all direction vectors perpendicular to the plane.
11. Let $C(\mathbf{c}, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{c}\| \leq r\}$ denote the ball including its boundary, called the closed ball. Show that a closed ball is a closed set.
12. Find all interior points of the set

$$A = (-1, 2) \setminus \left\{ x = \frac{1}{n} : n = 1, 2, 3, \dots \right\}.$$

Is this set open? Explain why or why not.

Lecture 2

Matrix Multiplication and Determinant

In the next four lectures, we study several important topics related to matrices. The central object of Calculus 2 is the derivative of a multivariable function, and the derivative of a multivariable function $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is an $m \times n$ matrix. In this sense, matrices are one of the core objects of Calculus 2. Many calculus textbooks do not treat matrix theory in sufficient depth, and as a result they are often forced to give explanations that miss the heart of the subject. Matrix theory is an essential part of the theory of derivatives of multivariable functions.¹

2.1 Notation

A matrix is written in rectangular form as follows:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

¹ **History Check Point:** The idea of representing the derivative of a multivariable function by a matrix was established gradually between the seventeenth and nineteenth centuries, together with the development of calculus and linear algebra. In the late seventeenth century, when Isaac Newton and Gottfried Wilhelm Leibniz introduced the concept of differentiation, their work dealt mainly with the slopes of tangent lines for functions of a single variable. In the eighteenth century, Leonhard Euler and Joseph-Louis Lagrange studied the change of multivariable functions in terms of partial derivatives and showed that the change of a function over a small region can be approximated by a linear expression. Through this work, the geometric ideas of multivariable differentiation, such as tangent planes and linear approximation, became firmly established.

In the nineteenth century, as linear algebra and matrix theory developed, the modern viewpoint of the derivative as a linear transformation emerged. In particular, Augustin-Louis Cauchy, in the 1820s, gave rigorous definitions of differentiation for functions of several variables, and Carl Gustav Jacobi, in the 1840s, organized partial derivatives into matrix form in his study of variable changes and function transformations. This led to what is now called the Jacobian matrix. Thus the modern definition was established: the derivative of a multivariable function is the linear transformation that best approximates the function, and the matrix representing that linear transformation is the derivative matrix, or Jacobian.

We call this an $m \times n$ matrix.² The expression $m \times n$ means that the matrix has m horizontal rows and n vertical columns. Remember that in an $m \times n$ matrix, the first number m indicates the number of rows and the second number n indicates the number of columns.

Let $\mathbb{R}^{m \times n}$ denote the set of all $m \times n$ matrices whose entries a_{ij} are real numbers. The entry at the intersection of the i -th row and the j -th column is called the (i, j) -entry and is denoted by a_{ij} . Here again, the first subscript i indicates the row number, and the second subscript j indicates the column number. As much as possible, we will use the subscript i for row numbers or for components of vectors in $\mathbb{R}^m$, so $1 \leq i \leq m$. Similarly, we will use the subscript j for column numbers or for components of vectors in $\mathbb{R}^n$, so $1 \leq j \leq n$. There will be situations in which this consistency cannot be maintained perfectly, but we will try to keep it whenever possible.

Let us first denote by $\tilde{\mathbf{a}}_i$ the vector consisting of the entries of the i -th row. This is a vector in $\mathbb{R}^n$ with n components. Since vectors are written as column vectors, we represent a row vector by attaching the transpose symbol:

$$\tilde{\mathbf{a}}_1^\top := [a_{11} \ a_{12} \ \cdots \ a_{1n}], \quad \cdots \quad, \tilde{\mathbf{a}}_m^\top := [a_{m1} \ a_{m2} \ \cdots \ a_{mn}].$$

From the row-vector point of view, the matrix A can be written as

$$A = \begin{bmatrix} \tilde{\mathbf{a}}_1^\top \\ \vdots \\ \tilde{\mathbf{a}}_m^\top \end{bmatrix}, \quad \tilde{\mathbf{a}}_i \in \mathbb{R}^n, \quad 1 \leq i \leq m.$$

It is sometimes convenient to think of each row as a single block. This point of view is especially useful when performing row operations.

We may also think of a matrix in terms of its columns. An $m \times n$ matrix has n columns. If we denote these column vectors by $\mathbf{a}_j$, then they are written as follows:

$$\mathbf{a}_1 := \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}, \quad \cdots \quad, \mathbf{a}_n := \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

From the column-vector point of view, the matrix can be written as

$$A = (\mathbf{a}_1, \cdots, \mathbf{a}_n), \quad \mathbf{a}_j \in \mathbb{R}^m, \quad 1 \leq j \leq n.$$

The column vectors lie in the image space. Therefore, writing a matrix in terms of its column vectors is especially useful when thinking about the image of a linear function. Applying the transpose symbol to a matrix, thereby interchanging rows and columns, produces the transpose matrix $A^\top$. Using the notation above, we may write

² In linear algebra, it is traditional to consider an $m \times n$ matrix. In other words, m is used for the number of rows and n for the number of columns. (Not $n \times m$.) This convention is related to the tradition of writing a vector-valued function in the form $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$.

$$A^\top = (\tilde{\mathbf{a}}_1, \dots, \tilde{\mathbf{a}}_m),$$

which is an $n \times m$ matrix. If a matrix A satisfies $A = A^\top$, then it is called symmetric. Therefore, for a matrix to be symmetric, it must first satisfy $m = n$, and such a matrix is called a square matrix.

Matrix Addition

We now introduce the basic operations on matrices. Forming a new matrix by adding two matrices A and B entry by entry is called matrix addition. In order to add two matrices, they must have the same size. For example, if both A and B are $m \times n$ matrices, then the entries of the sum $C = A + B$ are given by

$$c_{ij} = a_{ij} + b_{ij}.$$

Matrix subtraction is defined similarly: if $C = A - B$, then each entry is given by $c_{ij} = a_{ij} - b_{ij}$.

Scalar Multiplication

Multiplying a matrix by a real number is called scalar multiplication. The size of the matrix does not matter. If A is a matrix and x is a real number, then the matrix $B = xA$ is obtained by multiplying every entry a_{ij} by x . Thus each entry of B is given by

$$b_{ij} = xa_{ij}.$$

2.2 Matrix multiplication

Matrix multiplication produces a new matrix from two given matrices. The product of two matrices A and B is written simply as AB , without any symbol between them. The matrix product AB is defined when the number of columns of A is equal to the number of rows of B . In other words, if A is an $m \times n$ matrix and B is an $n \times \ell$ matrix, then the product is defined, and the resulting matrix AB is an $m \times \ell$ matrix. It is helpful to write and remember this in the form

$$(m \times n)(n \times \ell) = (m \times \ell).$$

Viewpoint #1: inner product

The matrix product $AB = C$ can be understood from several different points of view. The first viewpoint uses the inner product. The number of columns n of the matrix A is the length of each row vector of A , while the number of rows n of the matrix B is the length of each column

vector of B . Since these two vectors have the same length, we can take their inner product. The matrix whose entries are these inner products is an $m \times \ell$ matrix. Most textbooks use this as the definition of matrix multiplication.

Let A be a fixed $m \times n$ matrix and B a fixed $n \times \ell$ matrix. Then the product $C = AB$ is an $m \times \ell$ matrix. We have already used the notation $A = (a_{ij})$ for the entries of A . What notation should we choose for the entries of B and C ? Let us write $B = (b_{jk})$. Thus the subscript j , which represented the columns of A , is now used to represent the rows of B . Then we write $C = (c_{ik})$. The (i, k) -entry of the product C is the inner product of the i -th row vector of A and the k -th column vector of B . That is,

$$C = AB; \quad c_{ik} = \tilde{\mathbf{a}}_i^\top \mathbf{b}_k = (a_{i1}, \dots, a_{in}) \begin{bmatrix} b_{1k} \\ \vdots \\ b_{nk} \end{bmatrix} = \sum_{j=1}^n a_{ij} b_{jk}. \quad (2.1)$$

It is worth thinking about the logic behind this choice of indices. If the indices are chosen well, confusion is reduced. Practicing how to choose indices is also helpful in computer programming. One needs practice in order to use subscripts with consistency and clear rules.

Problem 2.1. Let

$$A = \begin{bmatrix} 2 & -1 & 3 & 0 \\ 4 & 0 & 1 & -2 \\ 1 & 5 & -1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \\ 1 & 4 \\ 5 & -1 \end{bmatrix}.$$

- (1) Write the vectors $\tilde{\mathbf{a}}_2$ and $\mathbf{b}_2$.
- (2) Write the vector $\tilde{\mathbf{a}}_3$.
- (3) Find the $(2, 2)$ -entry of the product AB .
- (4) Compute the matrix product $C = AB$.

Solution 2.1 We always regard vectors as column vectors.

$$\tilde{\mathbf{a}}_2 = \begin{bmatrix} 4 \\ 0 \\ 1 \\ -2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 2 \\ 0 \\ 4 \\ -1 \end{bmatrix}, \quad \tilde{\mathbf{a}}_3 = \begin{bmatrix} 1 \\ 5 \\ -1 \\ 3 \end{bmatrix}, \quad (AB)_{22} = 14.$$

One must perform matrix multiplication many times until it becomes natural. After enough practice, the left finger automatically moves along the rows and the right finger along the columns. $\square$

The inner product itself can also be understood as a matrix product: a row vector is a $1 \times n$ matrix and a column vector is an $n \times 1$ matrix, so their product has size

$$(1 \times n)(n \times 1) = 1 \times 1,$$

that is, a scalar. In other words, the inner product of two column vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ can be written as

$$\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a} \cdot \mathbf{b} = \mathbf{a}^T \mathbf{b} = [a_1 \cdots a_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \sum_{i=1}^n a_i b_i.$$

This point of view is useful when one studies the entries c_{ik} of a matrix product C one by one.

Commutative law of matrix multiplication?

Matrix multiplication does not satisfy the commutative law. In general,

$$AB \neq BA.$$

In fact, after reversing the order, the product may not even be defined. In order for both AB and BA to be defined, if A is an $m \times n$ matrix, then B must be an $n \times m$ matrix. But in that case, the product AB is an $m \times m$ matrix, while BA is an $n \times n$ matrix. Thus the only possibility for $AB = BA$ is when A and B are square matrices of the same size. Even then, however, the two products are generally not equal.

Problem 2.2. Check whether AB and BA are equal by multiplying the following square matrices:

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Solution 2.2 In fact, the case $AB = BA$ is relatively rare. However, since we are considering only simple 2×2 square matrices here and have included a few special cases, some of the matrices above do satisfy the equality.

First, the matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

consisting entirely of zeros is called the zero matrix. Whether one multiplies by it on the left or on the right, the result is always the zero matrix.

Entries such as a_{11}, a_{22}, a_{33} , whose row and column numbers are equal, are called diagonal entries. A matrix whose only possibly nonzero entries are diagonal entries is called a diagonal matrix. Among these, the $n \times n$ matrix whose diagonal entries are all 1 is called the identity matrix and is denoted by I_n . In the present example,

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is the identity matrix. This matrix returns A itself whether it is multiplied on the left or on the right.

If $AB = I_n$, then the matrix B is called the inverse matrix of A . In this case, one also has $BA = I_n$, so A is likewise the inverse matrix of B . In the above examples,

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$$

are inverses of each other. $\square$

Associative law of matrix multiplication

Matrix multiplication satisfies the associative law, and this fact is very important. The associative law means that the order of multiplication may be regrouped without changing the order of the matrices themselves. If three matrices A, B, C are given, then

$$(AB)C = A(BC)$$

holds. Here $(AB)C$ means that one first computes AB and then multiplies by C , whereas $A(BC)$ means that one first computes BC and then multiplies by A . The associative law of matrix multiplication is needed in order to understand the chain rule for multivariable functions, and its proof will be discussed in the next lecture.

Exercises

1. The following matrices are given:

$$A_1 = \begin{bmatrix} 1 & 4 & -2 \\ 0 & -3 & 5 \\ 2 & 1 & 0 \\ -1 & 2 & 3 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & -1 & 2 \\ 0 & 5 & 1 \\ -4 & 2 & 3 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 2 & 0 & 1 \\ 3 & -1 & 2 \\ 0 & 2 & 0 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 1 & 4 & -2 & 0 \\ 0 & -3 & 5 & 1 \\ 2 & 1 & 0 & 1 \end{bmatrix}$$

- (1) Find all pairs of matrices (A_i, A_j) with $i \neq j$ for which the matrix product is defined. How many such pairs are there in total?
- (2) Find all cases in which both matrix products $A_i A_j$ and $A_j A_i$ are defined. Among these, check whether there are any cases in which the equality $A_i A_j = A_j A_i$ holds.
- (3) How many products of three distinct matrices are defined? For one such case, check whether the associative law

$$(AB)C = A(BC)$$

holds.

2. Compute the determinant of each of the following matrices. These matrices will later be important in connection with changes of variables. (θ, ϕ, ρ, r , etc. are to be treated as constants.)

$$(1) \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix} \quad (3) \begin{bmatrix} \sin \phi \cos \theta & \rho \cos \phi \sin \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \cos \phi \cos \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\rho \sin \phi & 0 \end{bmatrix} \quad (3) \begin{bmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. Let $D = \begin{bmatrix} \cos \phi \cos \theta & -\sin \phi \sin \theta \\ \cos \phi \sin \theta & \sin \phi \cos \theta \\ -\sin \phi & 0 \end{bmatrix}$. Compute the determinant of $D^\top D$.
4. Let A and B be $10^6 \times 10^6$ matrices. Explain how one might compute the matrix product AB in parallel using 100 CPUs.